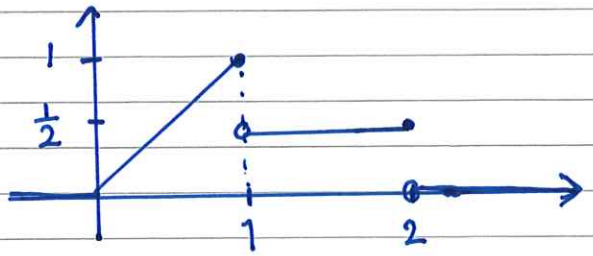
F78PA - December 2017 - Solutions.

1 (a) Graph of f_X:



f_X is a density because ① $f_X(x) \geq 0 \quad \forall x \in \mathbb{R}$

② $\int_{\mathbb{R}} f_X(x)dx$ = Area under graph of f_X

$= 1.$

(b) $\text{Im}X = \{x : f_X(x) > 0\} = (0,2]$ (Must explain answer).
(or $[0,2]$ is ok too).

$$E(X) = \int_{\mathbb{R}} x f_X(x)dx = \int_0^1 x \cdot x\,dx + \int_1^2 x \cdot \frac{1}{2}\,dx$$

$$= \frac{x^3}{3}\Big|_0^1 + \frac{x^2}{4}\Big|_1^2 = \frac{1}{3} + 1 - \frac{1}{4} = \frac{13}{12}.$$

(c) Since $\text{Im}X = (0,2]$, $F_X(x) = P(X \leq x) = 0$ if $x \leq 0$

and $F_X(x) = P(X \leq x) = 1$ if $x \geq 2$.

For $0 < x < 1$, $F_X(x) = P(X \leq x) = \int_0^x f_X(u)du = \int_0^x u\,du = \frac{x^2}{2}$.

For $1 \leq x < 2$, $F_X(x) = P(X \leq x)$

$$= \int_0^1 f_X(u)du + \int_1^x f_X(u)du.$$

$$= \int_0^1 u\,du + \int_1^x \frac{1}{2}\,du.$$

$$= \frac{u^2}{2}\Big|_0^1 + \frac{u}{2}\Big|_1^x = \frac{x}{2}$$

Summary: $F_X(x) = \begin{cases} 0 & x \leq 0 \\ \frac{x^2}{2} & 0 < x < 1 \\ \frac{x}{2} & 1 \leq x < 2 \\ 1 & x \geq 2. \end{cases}$

(Must be clear that the definition of a distribution function is known and understood).

F78PA - December 2017 - Solutions.

(d)(i) Since $\text{Im}X = [0,2]$, we know that $0 \le X \le 2$

and since $h(x) = e^x$ is an increasing function, we know that $0 \le X \le 2 \Rightarrow e^0 \le e^X \le e^2$, i.e. $1 \le Y \le e^2$ (since $Y = e^X$)

So $\text{Im}Y = [1, e^2]$,

For $1 \le y \le e^2$ (but not necessarily for other values of y!)

$$F_Y(y) = \underset{\text{def'n } F_Y}{} P(Y \le y) = P(e^X \le y) = P(\ln(e^X) \le \ln y)$$

(since $\ln$ is increasing)

$$= P(X \le \ln y)$$

$$= F_X(\ln y) \quad \text{def'n of } F_X.$$

(d)(ii) Since $\text{Im}Y = [1, e^2]$, we know that $f_Y(y) = 0$ for $y < 1$ and $y > e^2$.

For $1 \le y \le e^2$, by part (i), $F_Y(y) = F_X(\ln y)$

So: $f_Y(y) = F_Y'(y) = F_X'(\ln y) \cdot \frac{d}{dy}(\ln y)$ Chain Rule.

$$= f_X(\ln y) \cdot \frac{1}{y}.$$

There are two cases:

if $1 \le y \le e$, Then $\ln 1 \le \ln y \le \ln e$, i.e. $0 \le \ln y \le 1$

So: $f_Y(y) = f_X(\ln y) = \frac{\ln y}{y}$ since $f_X(x) = x$ when $0 \le x \le 1$.

If $e < y \le e^2$, Then $\ln e \le \ln y \le \ln e^2$, i.e. $1 \le \ln y \le 2$.

So: $f_Y(y) = f_X(\ln y) \cdot \frac{1}{y} = \frac{1}{2} \cdot \frac{1}{y}$ since $f_X(x) = \frac{1}{2}$ when $1 < x \le 2$.

F78PA - December 2017 - Solutions

(2) X = minutes after 11 pm when Fred arrives. $X \sim U(0,60)$.

Trains depart at 11.45 pm and midnight.

(a) T = length of time in minutes that Fred has to wait for the next train.

(i) $P(T \le 10) = P(35 \le X \le 45) + P(50 \le X \le 60)$

$= \frac{10}{60} + \frac{10}{60} = \frac{1}{3}$ (since $X \sim U(0,60)$).

$P(T \le 30) = P(15 \le X \le 60)$ (since $T \le 30 \Leftrightarrow$ Fred arrives after 11.15)

$= \frac{45}{60}$ since $X \sim U(0,60)$

(ii) $0 \le T \le 45$ since the longest that Fred will have to wait for a train is 45 min (if he arrives at 11pm).

So $F_T(t) = P(T \le t) = 0$ if $t < 0$

$F_T(t) = P(T \le t) = 1$ if $t \ge 45$.

For $0 \le t < 15$, $F_T(t) = P(T \le t) = P(45-t \le X \le 45)$

$+ P(60-t \le X \le 60)$

(since $X \sim U(0,60)$) $= \frac{t}{60} + \frac{t}{60} = \frac{t}{30}$

For $15 \le t \le 45$, $F_T(t) = P(T \le t) = P(45-t \le X \le 60)$

(since $X \sim U(0,60)$) $= \frac{15+t}{60} = \frac{1}{4} + \frac{t}{60}$

(Answer must demonstrate understanding).

(iii) From (ii) we obtain $f_T(t) = F_T'(t) = \begin{cases} \frac{1}{30} & 0 \le t < 15 \\ \frac{1}{60} & 15 \le t < 45 \\ 0 & \text{o.w.} \end{cases}$

So $E(T) = \int_0^{45} t f_T(t)\,dt = \int_0^{15} \frac{t}{30}\,dt + \int_{15}^{45} \frac{t}{60}\,dt = \frac{2250}{120} = 18.75$ min.

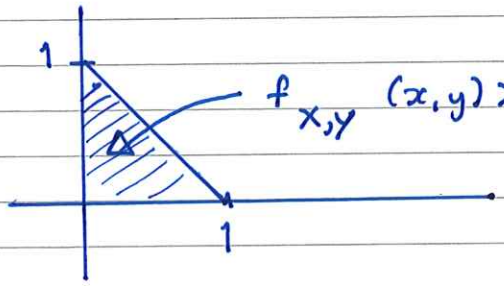
F78PA - December 2017 - Solutions.

(b) $P(Y=10) = P(0 \le X \le 45) = \frac{45}{60} = \frac{3}{4}$ (since $X \sim U(0,60)$)

$P(Y=15) = P(45 < X \le 60) = \frac{15}{60} = \frac{1}{4}$

So $E(Y) = 10 \cdot P(Y=10) + 15\,P(Y=15) = 10 \times \frac{3}{4} + 15 \times \frac{1}{4} = \frac{45}{4} = £11.25$.

3(a).



$f_{X,Y}(x,y) > 0$ in this region.

(b) $f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy = 0$ if $x<0$ or $x>1$

because $f_{X,Y}(x,y) = 0$ when $x<0$ or $x>1$

(i.e. since $f_{X,Y}(x,y) = 0$ outside the shaded region)

For $0 \le x \le 1$

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy = \int_0^{1-x} 12y^2\,dy = 4(1-x)^3$$

(c)(i) For $0 \le x,y$ s.t. $0 \le x+y \le 1$ (and $x \ne 1$)

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{12y^2}{4(1-x)^3} = \frac{3y^2}{(1-x)^3}$$

AND $f_{Y|X}(y|x) = 0$ otherwise.

(c)(ii) $P(Y \le \frac{1}{4} \mid X = \frac{1}{2}) = \int_0^{1/4} f_{Y|X}(y|\tfrac{1}{2})\,dy = \int_0^{1/4} 24y^2\,dy = \frac{1}{8}$.

(iii) $E(Y \mid X = \frac{1}{2}) = \int_0^{1/2} y f_{Y|X}(y|\tfrac{1}{2})\,dy = \int_0^{1/2} 24y^3\,dy = \frac{3}{8}$.

(d) Since $f_{X,Y}(x,y) \ne 0$ only when $0 \le x+y \le 1$ (and $x,y>0$)

we must have $0 \le X+Y \le 1$, i.e. $0 \le Z \le 1 \Rightarrow \mathrm{Im}\,Z = [0,1]$.

F78PA - December 2017 - Solutions

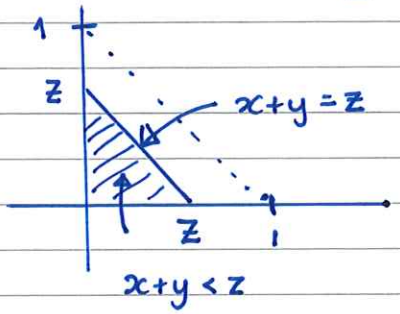
(d)(i) continued.

Since $\text{Im}\, Z = [0,1]$, we have $F_Z(z) = P(Z \le z) = 0$ if $z < 0$

and $F_Z(z) = P(Z \le z) = 1$ if $z \ge 1$.

For $0 \le z < 1$, $F_Z(z) = P(Z \le z)$

$= P(X+Y \le z)$ by def'n of Z and F_Z.



$= \int_0^z \int_0^{z-x} f_{X,Y}(x,y)\, dy\, dx.$

$= \int_0^z \int_0^{z-x} 12y^2\, dy\, dx.$

$= \int_0^z 4(z-x)^3\, dx$

$= -(z-x)^4 \Big|_0^z = z^4.$

(ii) $P(Z \le \frac{1}{2} \mid Z > \frac{1}{2}) = \dfrac{P(\frac{1}{4} < Z \le \frac{1}{2})}{P(Z > \frac{1}{4})} = \dfrac{F_Z(\frac{1}{2}) - F_Z(\frac{1}{4})}{1 - F_Z(\frac{1}{4})} = \dfrac{\frac{1}{16} - \frac{1}{256}}{1 - \frac{1}{256}} = \dfrac{1}{17}$

4(a) $M = \max\{X_1, X_2, \ldots, X_n\}$.

$F_M(x) = P(M \le x)$ (by def'n of F_M)

$= P(\max\{X_1, \ldots, X_n\} \le x)$ by def'n of M.

$= P(X_1 \le x, \ldots, X_n \le x)$ since $\max\{X_1 \ldots X_n\} \le x$ iff $X_i \le x \ \forall i$

by indep. $= P(X_1 \le x) P(X_2 \le x) \cdots P(X_n \le x)$

$= F_{X_1}(x) F_{X_2}(x) \cdots F_{X_n}(x)$ by def'n of F_{X_i}

$= (F_X(x))^n$ since all the X's have the same df, F_X.

(b)(i) Since $T = \max\{X_A, X_B\}$ and X_A and X_B are indep $\&$ the same df,

part (a) $\Rightarrow F_T(t) = (F_X(t))^2 = \begin{cases} 0 & t < 0 \\ t^4 & 0 \le t \le 1 \\ 1 & 1 \le t \end{cases}$

F78PA December 2017 - Solutions.

④ (b) (i) ctd. $f_T(t) = F_T'(t) = \begin{cases} 4t^3 & 0<t<1 \\ 0 & \text{ow.} \end{cases}$

(ii) Since the conditional distribution of Y given $T=t$ is $U(t+1, t+2)$ for $0<t<1$, the conditional expectation of Y given $T=t$ is: $E(Y|T=t) = \frac{(t+1)+(t+2)}{2}$ ← expected value of $U(t+1, t+2)$

NB It is incorrect to say that Y is $U(t+1, t+2)$.

(iii) Partition Theorem $\Rightarrow E(Y) = \int_0^1 E(Y|T=t) \cdot f_T(t)\,dt$

$$= \int_0^1 \left(t + \tfrac{3}{2}\right) \cdot 4t^3\,dt$$

$$= \int_0^1 4t^4 + 6t^3\,dt = \frac{23}{10} = 2.3 \text{ hours.}$$

5 (a) $M_X(\theta) = E(e^{\theta X})$ def'n of $M_X(\theta)$

$$= \int_1^\infty e^{\theta x} f_X(x)\,dx.$$

$$= \int_1^\infty e^{\theta x} \cdot e^{-x+1}\,dx.$$

$$= e^1 \int_1^\infty e^{-(1-\theta)x}\,dx$$ ← this will be finite iff $1-\theta>0$ i.e. $\theta<1$.

$$= e\left(\frac{-e^{-(1-\theta)x}}{(1-\theta)}\Big|_1^\infty\right)$$

$$= e\left(\frac{-e^{-(1-\theta)}}{1-\theta}\right) = \frac{e^\theta}{1-\theta}$$

$$E(X) = M_X'(0) = \frac{d}{d\theta}\left(\frac{e^\theta}{1-\theta}\right)\Big|_{\theta=0} = \frac{e^\theta + e^\theta}{(1-\theta)^2}\Big|_{\theta=0} = 2$$

(or use the fact that $M_X(\theta) = 1 + E(X)\theta + \frac{E(X^2)}{2!}\theta^2 + \dots$

and $M_X(\theta) = \frac{e^\theta}{1-\theta} = \left(1+\theta+\frac{\theta^2}{2!}+\dots\right)(1+\theta+\dots)$

and equate coefficients).

F78 PA - December 2017 - Solutions

(b) $M_Y(\theta) = E(e^{\theta Y}) = E(e^{\theta(X-1)})$ since $Y = X-1$

$\overset{\text{def'n } M_Y}{=} e^{-\theta} E(e^{\theta X})$ properties of E.

$= e^{-\theta} M_X(\theta)$ (def'n of E M_X

$= e^{-\theta} \cdot \frac{e^{\theta}}{1-\theta} = \frac{1}{1-\theta}$ ← mgf for Exp(1)

∴ By the uniqueness of mgf, $Y \sim Exp(1)$.

6(a) $M_N(\theta) = E(e^{\theta N}) = E(e^{\theta(X_1+\cdots+X_{52})})$ by def'n of M_N and N

$= E(e^{\theta X_1})E(e^{\theta X_2})\cdots E(e^{\theta X_{52}})$ since X_i's are indep.

$= (M_{X_1}(\theta))^{52}$ def'n of M_{X_1} and since X_i's have same dist (and hence, same mgf)

$= (e^{-1+e^{\theta}})^{52}$ since $X \sim Po(1)$ and mgf $Po(1)$ is $e^{-1+e^{\theta}}$

$= e^{-52+52e^{\theta}}$ ← mgf for Poisson(52).

So, by the uniqueness of mgf, $N \sim Po(52)$.

(b) Central Limit Theorem: Suppose $X_1, X_2, X_3, \ldots$ are iid with $E(X) = \mu$ and $Var(X) = \sigma^2 < \infty$.

Then for any $-\infty \le a < b < \infty$,

$$\lim_{n\to\infty} P\left(a \le \frac{X_1+\cdots+X_n - n\mu}{\sqrt{n\sigma^2}} \le b\right) = P(a \le Z \le b) \text{ where } Z \sim N(0,1).$$

$P(N > 40) = P(X_1+\cdots+X_{52} > 40) = P(X_1+\cdots+X_{52} \ge 40.5)$ continuity correction since N is discrete

def'n of N

$$= P\left(\frac{X_1+\cdots+X_{52} - 52(1)}{\sqrt{52(1)}} \ge \frac{40.5 - 52(1)}{\sqrt{52(1)}}\right)$$

$\approx P(Z \ge -1.59) = P(Z \le 1.59)$ (by symmetry of the distribution of Z)

by Central Limit Theorem.

$= 0.9441.$